



The diagram illustrates a cryogenic distillation process for separating ethane and propylene. The process begins with a **Feed** entering an **Inlet Separator**. The liquid from the inlet separator goes to the bottom of a **Stabilizer** column, which has 10 stages. The top product of the stabilizer is **Overhead Vapour**, which is condensed in a **Condenser** and collected in a **Reflux Drum**. The liquid from the reflux drum is pumped back to the top of the stabilizer column. The bottom product of the stabilizer is **Bottoms Liquid**, which is pumped to a **Reboiler** and then back to the bottom of the stabilizer column. The overhead vapour is also used for **Vapour Pressure Control**. The gas from the top of the inlet separator goes to a **Gas/Gas Exchanger**, which is part of a **Dew Point Control** system. The gas from the gas/gas exchanger goes to a **Chiller**, which is also part of the dew point control system. The chiller is connected to a **Refrigeration** unit. The gas from the chiller goes to a **Low Temperature Separator**, which is also part of the dew point control system. The liquid from the low temperature separator goes to the bottom of the inlet separator. The gas from the low temperature separator goes to the **Gas/Gas Exchanger**. The gas from the gas/gas exchanger is the **Sales Gas**.

The vapour from a feed stream is passed through a gas to gas exchanger and then a chiller, reducing the temperature to condense out heavier components. The vapour from the low temperature separator is then reheated in the gas to gas exchanger to become the sales gas, while pre-cooling the inlet separator vapour. The dew point of the sales gas is checked against a target value and the temperature of the low temperature separator is adjusted until that target is met.

The feed conditions are known and given pressure drops for the gas to gas exchanger and chiller all that is needed is an initial estimate for the low temperature separator temperature to define it. A temperature approach between the inlet separator vapour and the sales gas allows the sale gas to be completely determined. With three of the four gas/gas stream known, a heat balance allows the temperature to the chiller to be determined.

The liquids from the inlet and low temperature separators are fed to a stabilizer. A solver adjusts the overhead flow rate to meet a desired vapour pressure in the bottoms product.

DewPtTarget	-5 degC
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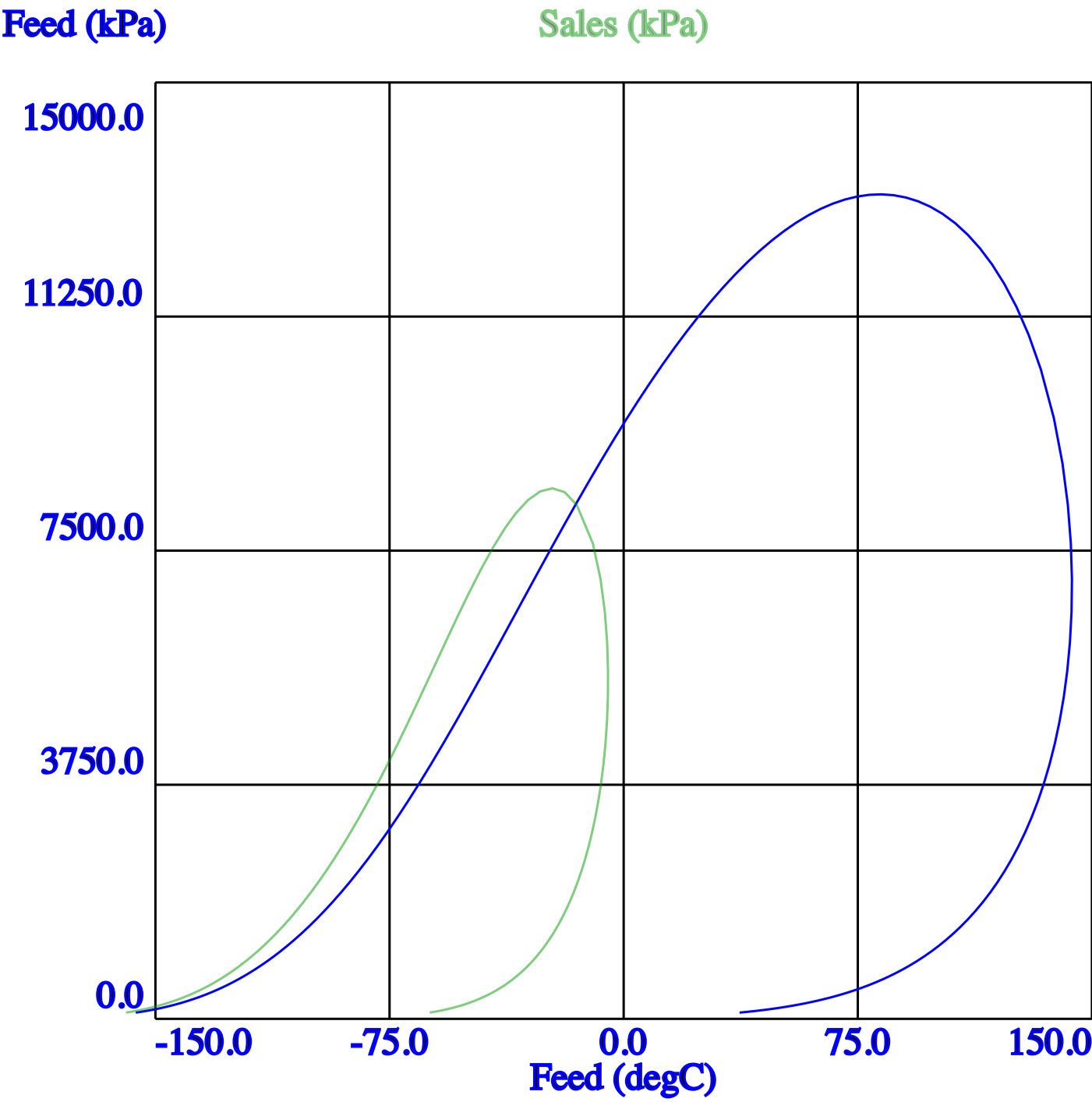
DewPtSolver

feed

Label	Unit	B	V	L
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q	Fraction	0.56289	1.00000	0.00000
t	degC	15.00	15.00	15.00
p	kPa	3000.00	3000.00	3000.00
f	kmol/h	4980.22	2803.30	2176.92
h	kJ/kmol	-10266.73	-1405.26	-21677.96
s	kJ/kmol-K	-42.08	-25.31	-63.67
hflow	W	-14202939.06307	-1094264.58353	-13108674.47954
x	Methane	0.49296	0.77051	0.13555
x	Ethane	0.14085	0.14872	0.13071
x	Propane	0.07042	0.03947	0.11028
x	Isobutane	0.06338	0.01822	0.12154
x	n-Butane	0.05634	0.01299	0.11216
x	Isopentane	0.04930	5.108849e-3	0.10620
x	n-Pentane	0.04225	3.617859e-3	0.09201
x	n-Hexane	0.03521	9.973211e-4	0.07927
x	n-Heptane	0.02817	2.907505e-4	0.06407
x	n-Octane	0.02113	8.078257e-5	0.04823
dmass	kg/m^3	84.65165	30.20224	571.69592

Envelopes



lts

Label	Unit	B	V	L
q	Fraction	0.95707	1.00000	0.00000
t	degC	-10.71	-10.71	-10.71
p	kPa	2862.10	2862.10	2862.10
f	kmol/h	2803.30	2682.95	120.35

h	kJ/kmol	-3192.97	-2435.87	-20071.72
s	kJ/kmol-K	-31.50	-29.79	-69.61
hflow	W	-2486349.34507	-1815366.31914	-670983.02593
x	Methane	0.77051	0.79730	0.17326
x	Ethane	0.14872	0.14666	0.19459
x	Propane	0.03947	0.03370	0.16805
x	Isobutane	0.01822	0.01189	0.15933
x	n-Butane	0.01299	7.612979e-3	0.13290
x	Isopentane	5.108849e-3	1.691554e-3	0.08129
x	n-Pentane	3.617859e-3	1.041846e-3	0.06105
x	n-Hexane	9.973211e-4	9.562803e-5	0.02110
x	n-Heptane	2.907505e-4	9.275094e-6	6.565903e-3
x	n-Octane	8.078257e-5	8.518498e-7	1.862741e-3
dmass	kg/m ³	33.69427	30.72607	542.70622

SalesGas

Label	Unit	B	V	L
q	Fraction	1.00000	1.00000	0.00000
t	degC	10.00	10.00	10.00
p	kPa	2793.16	2793.16	2793.16
f	kmol/h	2682.95	2682.95	0.00
h	kJ/kmol	-1458.42	-1458.42	-1458.42
s	kJ/kmol-K	-26.03	-26.03	-26.03
hflow	W	-1086905.84430	-1086905.84430	0.00000
x	Methane	0.79730	0.79730	0.26778
x	Ethane	0.14666	0.14666	0.22283
x	Propane	0.03370	0.03370	0.15866
x	Isobutane	0.01189	0.01189	0.13281
x	n-Butane	7.612979e-3	7.612979e-3	0.10430
x	Isopentane	1.691554e-3	1.691554e-3	0.05737
x	n-Pentane	1.041846e-3	1.041846e-3	0.04075
x	n-Hexane	9.562803e-5	9.562803e-5	0.01158
x	n-Heptane	9.275094e-6	9.275094e-6	3.150459e-3
x	n-Octane	8.518498e-7	8.518498e-7	7.599288e-4

dmass	kg/m^3	26.66304	26.66304	26.66304
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ChillerDuty	663.62 kW
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refrigPower	268.36 kW
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VPSolver

x (solved)	fx
0.08312	-1.080682e-4

Stab

Stage	T degC	P kPa	V kmol/h	L kmol/h	Q kW	Feeds kmol/h	Draws kmol/h
1	66.43824	1300.00000		3233.24931	-19510.17038		V 1616.62477
2	92.90975	1305.55556	4849.87408	3483.14543	0.00000		
3	102.60425	1311.11111	5099.77020	3618.21635	0.00000	120.34538	
4	109.56951	1316.66667	5114.49574	3534.79278	0.00000		
5	115.87587	1322.22222	5031.07217	6460.41800	0.00000	2176.92153	
6	131.62998	1327.77778	5779.77586	6983.88004	0.00000		
7	139.33799	1333.33333	6303.23790	6981.57949	0.00000		
8	146.89424	1338.88889	6300.93735	6721.58182	0.00000		
9	157.71049	1344.44444	6040.93968	6265.61716	0.00000		
10	174.24697	1350.00000	5584.97502		34742.64361		L 680.64214